

Androgens

It has long been recognized that sebaceous glands require androgenic stimulation to produce significant quantities of sebum. Individuals with a genetic deficiency of androgen receptors (complete androgen insensitivity) have no detectable sebum secretion and do not develop acne. However, there is still a question as to which androgen is physiologically significant. Although the most powerful androgens are testosterone and its end-organ reduction product dihydrotestosterone, levels of testosterone do not parallel the patterns of sebaceous gland activity. For example, testosterone levels are many-fold higher in males than in females, with no overlap between the sexes, whereas average rates of sebum secretion are only slightly higher in males than in females, with considerable overlap between the groups. Also, sebum secretion starts to increase in children during adrenarche, a developmental event that precedes puberty by approximately 2 years.

The weak adrenal androgen dehydroepiandrosterone sulfate (DHEAS) might be a significant regulator of sebaceous gland activity through its conversion to testosterone and dihydrotestosterone in the sebaceous gland. Levels of DHEAS are high in newborns, are very low in 2- to 4-year-old children, and start to rise when sebum secretion starts to increase. In adulthood, DHEAS levels show considerable individual variation but are only slightly higher in men than in women on the average. There is a decline in DHEAS levels in both sexes starting in early adulthood and continuing throughout life. This decline parallels the decline of sebum secretion. DHEAS is present in the blood in high concentration. The enzymes required to convert DHEAS to more potent androgens are present in sebaceous glands. These include 3β -hydroxysteroid dehydrogenase, 17β -hydroxysteroid dehydrogenase, and 5α -reductase. Each of these enzymes exists in two or more isoforms that exhibit tissue-specific differences in their expression. The predominant isozymes in the sebaceous gland include type 1 3β -hydroxysteroid dehydrogenase, type 2 17β -hydroxysteroid dehydrogenase, and type 1 5α -reductase.

Retinoids

Isotretinoin (13-cis retinoic acid) is the most potent pharmacologic inhibitor of sebum secretion. Significant reductions in sebum production can be observed as early as 2 weeks after use. Histologically, sebaceous glands are markedly reduced in size, and individual sebocytes appear undifferentiated, lacking the characteristic cytoplasmic accumulation of sebaceous lipids.

Isotretinoin does not interact with any of the known retinoid receptors. It may serve as a prodrug for the synthesis of all-trans retinoic acid or 9-cis retinoic acid, which do interact with retinoid receptors; however, it has greater sebosuppressive action than all-trans or 9-cis retinoic acid. The mechanism by which 13-cis retinoic acid lowers sebum secretion is unknown, but it inhibits the α -hydroxysteroid activity of retinol dehydrogenase and leads to decreased androgen synthesis. In addition, isotretinoin triggers cell cycle arrest in human sebocytes and immortalized cell culture models of human sebocytes (SZ95 and SEB-1) and also induces apoptosis in SEB-1 sebocytes. Inhibition of androgen synthesis, cell cycle arrest, and apoptosis by 13-cis retinoic acid may explain the reduction of sebaceous gland size after treatment.

Melanocortins

Melanocortins include melanocyte-stimulating hormone and adrenocorticotrophic hormone. In rodents, melanocortins increase sebum production. Transgenic mice deficient in the melanocortin-5 receptor have hypoplastic sebaceous glands and reduced sebum production. The melanocortin-5 receptor has been identified in human sebaceous glands, where it may play a role in the modulation of sebum production.

Peroxisome Proliferator-Activated Receptors

PPARs are orphan nuclear receptors that are similar to retinoid receptors in many ways.